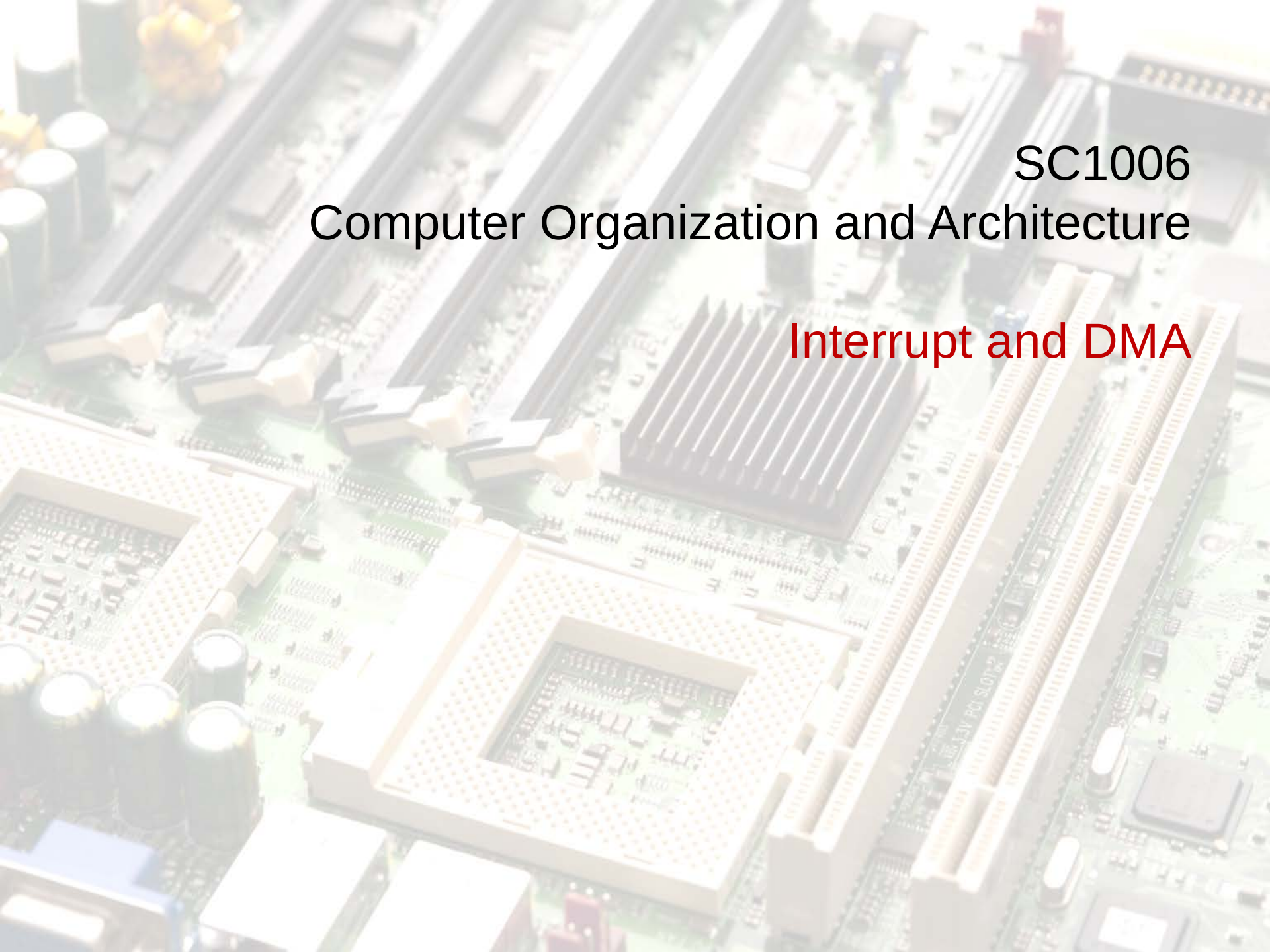




SC1006

Computer Organization and Architecture

Interrupt and DMA

Interrupt Triggered Data Transfer

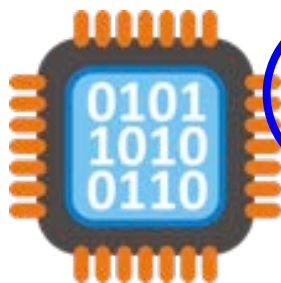
Main Routine

```
main  MOV R0, #0
      MOV R2, #8

Loop  ADD R1,R1,R2
      BEQ skip
      ADD R0,R0,R2

skip  .....
```

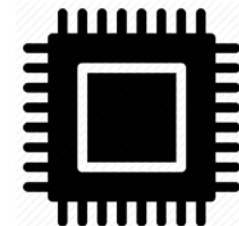
I/O Device #3
Interrupt Service
Routine



INTERRUPT REQUEST

DATA TRANSFER

I/O Device #3



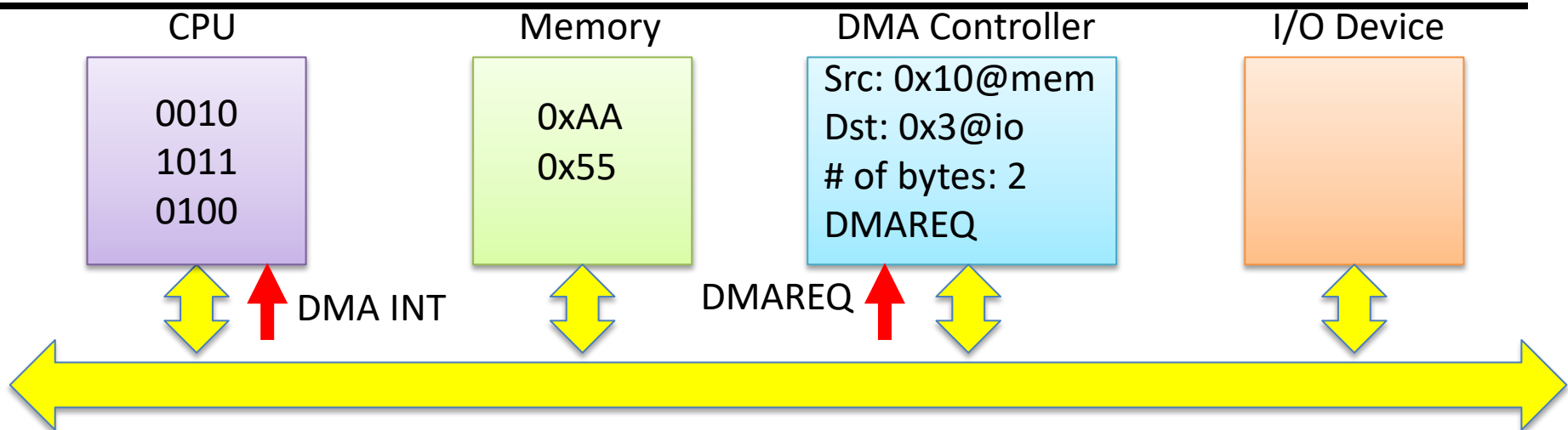
Interrupt Vector Table

Interrupt Vector Table

0	Reset ISR Addr
1	...
2	...
3	...
4	...
5	...
6	...
7	Ext INT 3 ISR Addr
8	SPI ISR Addr
9	UART ISR Addr
A	

- How does the CPU know the **location** of the corresponding interrupt service routine for each interrupts?
- The **starting address of each interrupt** is stored in a table known as the **interrupt vector table**
- Each interrupt has a **unique index to the vector table**, e.g. UART interrupt could be in index 9, so if a UART interrupt occurs, CPU will know where to branch to by checking index 9 in the vector table.
- The values and interrupt source in the table on the left are for **illustration purpose only**, different processor has different indexing for their interrupts.

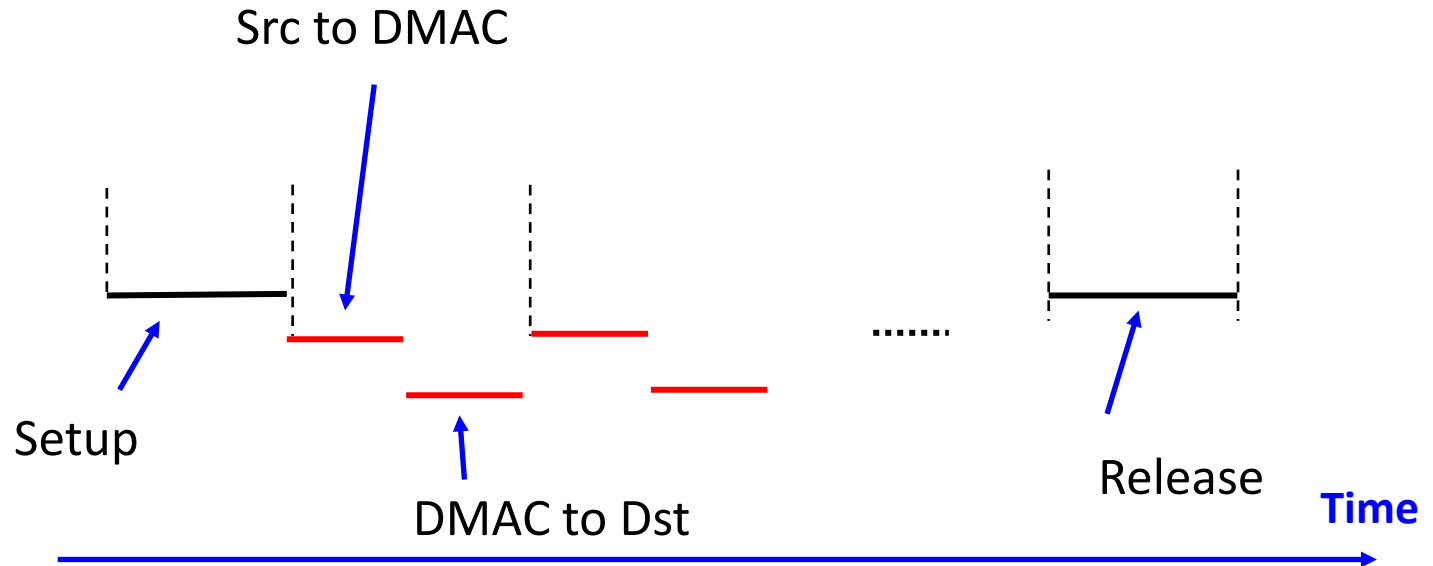
Basic DMA Process



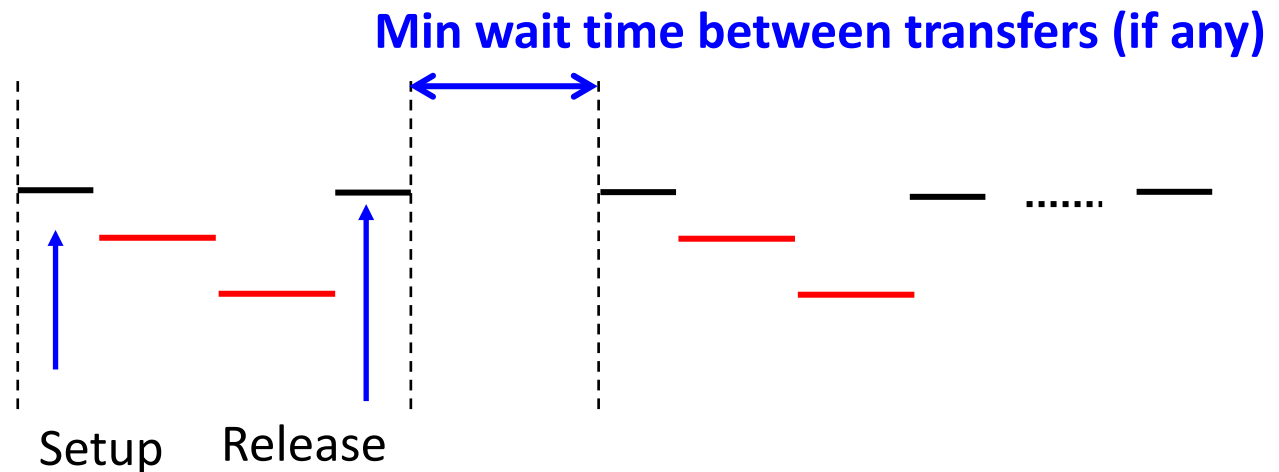
- DMA Configuration of Parameters (CPU to DMAC)
 - Source Address
 - Destination Address
 - Amount of data to transfer
- CPU proceeds with its own task while DMAC wait for DMA Trigger Signal (DMAREQ).
- Once DMA Trigger occurs, DMAC request to take over system bus.
- DMAC transfer data according to configuration.
- DMAC notify CPU of data completion (typically via interrupts) and release system bus.

Fetch and Deposit (Timings)

Burst



Cycle Stealing



3 Basic DMA Mode of Operation

- Burst
- Cycle Stealing
- Transparent

Thinking questions toward understanding...

1. We have entered the digital era. Why do we still use analog-based/related sensors, amplifiers, filters, converters (ADC, DAC), etc?

2. In which situation(s), would you like to select

- Polling
- Interrupt
- DMA

for data transfers?

